

Solution to Ramanujan Challenge Problem 2.5: Efficient Rational Approximation of Catalan's Constant G

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Abstract

We prove, for every one of the three terminal columns, that the matrix quotient in Problem 2.5 tends to Catalan's constant. The proof is exact. A Meijer- G trajectory is related to the printed matrix by an explicit inverse-transpose identity. A positive fourfold integral identifies the unique recessive solution, and an exact Barnes-moment calculation evaluates its initial vector. We also give the exact inverse Delannoy transform and the requested finite integral formula for its coefficient error. No finite-prefix guess, decimal recognition, or non-effective sign argument is used.

1 The matrix and the assertion

Let $M(n) = (m_{ij}(n))_{1 \leq i, j \leq 3}$ be the matrix in the question. For reference, its entries are

$$\begin{aligned} m_{11} &= (-2n - 5)(n + 3)^2(136n^4 + 1424n^3 + 5548n^2 + 9551n + 6141), \\ m_{12} &= 384n^6 + 6384n^5 + 44168n^4 + 162698n^3 + 336377n^2 + 369933n + 169011, \\ m_{13} &= -480n^4 - 4980n^3 - 19210n^2 - 32690n - 20730, \\ m_{21} &= (n + 2)^2(n + 3)^2(4n + 10)(48n^3 + 386n^2 + 1017n + 879), \\ m_{22} &= (n + 2)^2(-272n^5 - 3848n^4 - 21732n^3 - 61184n^2 - 85761n - 47808), \\ m_{23} &= (n + 2)^2(320n^3 + 2540n^2 + 6610n + 5640), \\ m_{31} &= (-4n - 10)(n + 2)^2(n + 3)^2(32n^4 + 302n^3 + 1037n^2 + 1530n + 813), \\ m_{32} &= (n + 2)^2(192n^6 + 2984n^5 + 19116n^4 + 64452n^3 + 120256n^2 + 117279n + 46476), \\ m_{33} &= (n + 2)^2(-16n^5 - 408n^4 - 2912n^3 - 8884n^2 - 12254n - 6240). \end{aligned}$$

Put

$$\mathcal{M}_N = M(0)M(1) \cdots M(N - 1), \quad A = \begin{pmatrix} 30921 & -32972 & 8240 \\ 33750 & -36000 & 9000 \end{pmatrix},$$

where the empty product is the identity, and define

$$A\mathcal{M}_N = \begin{pmatrix} P_{N,1} & P_{N,2} & P_{N,3} \\ Q_{N,1} & Q_{N,2} & Q_{N,3} \end{pmatrix}.$$

Finally,

$$G = \sum_{r=0}^{\infty} \frac{(-1)^r}{(2r + 1)^2}.$$

Theorem 1 (Main assertion). *For $j = 1, 2, 3$,*

$$\lim_{N \rightarrow \infty} \frac{P_{N,j}}{Q_{N,j}} = G.$$

The proof is completed in Theorem 19.

Lemma 2 (Catalan moment). *The improper integral is absolutely convergent and*

$$G = \int_0^1 \frac{-\log t}{1+t^2} dt, \quad \int_0^1 (-\log t) dt = 1.$$

Proof. The second identity follows by integration by parts. For $0 \leq t < 1$,

$$\frac{1}{1+t^2} = \sum_{r \geq 0} (-1)^r t^{2r}.$$

The partial sums have absolute value at most 2, while $-\log t$ is integrable on $(0, 1)$. Dominated convergence and $\int_0^1 t^{2r} (-\log t) dt = (2r+1)^{-2}$ give the first identity. $\square$

The following exact moment identity will also be used in the Delannoy appendix. It is not used to infer decay merely from a finite alternating sum.

Lemma 3 (Exact moment lift). *Let $\mathbf{p}_0$ and $\mathbf{q}_0$ be the two rows of A , set*

$$\mathbf{R}_0(X) = \mathbf{q}_0 - (1+X)\mathbf{p}_0, \quad R_{N,j}(X) = (\mathbf{R}_0(X)\mathcal{M}_N)_j.$$

Then $R_{N,j}(X) = Q_{N,j} - (1+X)P_{N,j}$ and

$$Q_{N,j} = \int_0^1 (10-18x)R_{N,j}(x) dx, \tag{1}$$

$$P_{N,j} = \int_0^1 (6-12x)R_{N,j}(x) dx, \tag{2}$$

$$GQ_{N,j} - P_{N,j} = \int_0^1 \frac{-\log t}{1+t^2} R_{N,j}(t^2) dt. \tag{3}$$

Proof. The displayed affine formula for $R_{N,j}$ follows directly from the definition. If $R(X) = a+bX$, then the four elementary moments

$$\int_0^1 (10-18x) dx = 1, \quad \int_0^1 x(10-18x) dx = -1, \quad \int_0^1 (6-12x) dx = 0, \quad \int_0^1 x(6-12x) dx = -1$$

give (1) and (2). Lemma 2 gives

$$\int_0^1 \frac{-\log t}{1+t^2} (Q - (1+t^2)P) dt = GQ - P,$$

which is (3). The logarithmic endpoint is integrable, so these are ordinary improper Lebesgue integrals. $\square$

2 Inverse-transpose selection

We record the standard form of the recessive-solution principle needed below. It is a consequence of the Poincaré–Perron theorem in its higher-order (discrete Levinson) form; see [1].

Proposition 4 (Recessive recognition and Miller selection). *Suppose a row companion system $X_{n+1} = X_n C(n)$ satisfies $C(n) = C_\infty + O(n^{-1})$, where C_∞ is invertible and diagonalizable. Suppose C_∞ has a simple eigenvalue μ such that*

$$0 < |\mu| < \min\{|\lambda| : \lambda \in \text{spec}(C_\infty), \lambda \neq \mu\}.$$

If a nonzero scalar solution F_n has $\limsup |F_n|^{1/n} \leq |\mu|$, it spans the unique recessive solution line. Writing

$$\Phi_C(N) = C(0) \cdots C(N-1), \quad \mathbf{F}_0 = (F_0, \dots, F_{r-1})^T,$$

one has

$$[\Phi_C(N)^{-T} w_N] \longrightarrow [\mathbf{F}_0]$$

for every terminal vector w_N whose coefficient in the corresponding dominant dual direction is nonzero.

Moreover, if $T(n)U(n+1) = U(n)C(n)$ with rational invertible $U(n)$, then

$$[\Phi_T(N)^{-T} e_j] \longrightarrow [U(0)^{-T} \mathbf{F}_0] \tag{4}$$

whenever $U(N)^T e_j$ has nonzero recessive-dual connection.

Proof. The spectral gap splits the asymptotic solution space into the one-dimensional μ -space and its complementary block. Discrete Levinson asymptotics show that every solution outside the first space has strictly larger exponential rate. This proves the recognition statement. Inverting and transposing a fundamental product replaces each exponential factor λ^N by λ^{-N} ; hence the recessive primal line becomes the unique dominant dual line. This proves the projective assertion.

Finally, multiplication of the coboundary identities gives

$$\Phi_T(N) = U(0)\Phi_C(N)U(N)^{-1}, \quad \Phi_T(N)^{-T} = U(0)^{-T}\Phi_C(N)^{-T}U(N)^T.$$

Rational gauges have at most polynomial growth along the positive integers and therefore do not alter an exponentially isolated line. This gives (4). $\square$

3 An exact Meijer–G trajectory

We use the Barnes convention of [2, Sec. 16.17],

$$G_{p,q}^{m,r} \left(z \left| \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix} \right. \right) = \frac{1}{2\pi i} \int_L \frac{\prod_{j=1}^m \Gamma(b_j - s) \prod_{j=1}^r \Gamma(1 - a_j + s)}{\prod_{j=m+1}^q \Gamma(1 - b_j + s) \prod_{j=r+1}^p \Gamma(a_j - s)} z^s ds.$$

For $n \geq 0$ define

$$F_n(z) = G_{4,4}^{4,2} \left(z \left| \begin{matrix} -1-n, -1-n, n+\frac{5}{2}, n+3 \\ \frac{3}{2}, \frac{1}{2}, 1, 0 \end{matrix} \right. \right), \quad F_n = F_n(1), \tag{5}$$

and set $\Theta = z d/dz$ and

$$Y_n = (F_n, (\Theta F_n)(1), (\Theta^2 F_n)(1)).$$

Lemma 5 (Differential reduction). *At $z = 1$,*

$$(\Theta^3 F_n)(1) = -\frac{(n+2)^3(2n+3)}{7}F_n + \frac{2n^2+8n+5}{14}(\Theta F_n)(1) + \frac{8n^2+30n+39}{14}(\Theta^2 F_n)(1). \quad (6)$$

Proof. The Meijer- G differential equation for (5) is

$$[z(\Theta + n + 2)^2(\Theta - n - \frac{3}{2})(\Theta - n - 2) - (\Theta - \frac{3}{2})(\Theta - \frac{1}{2})(\Theta - 1)\Theta] F_n(z) = 0.$$

At $z = 1$ the fourth-order terms cancel. Expansion and division by $7/2$ gives (6). $\square$

Define $T(n) = (t_{ij}(n))$ by

$$\begin{pmatrix} \frac{4(n+2)(17n^3+111n^2+240n+171)}{(n+1)(n+3)(2n+3)(2n+5)} & \frac{(n+2)(24n^2+101n+102)}{(n+1)(2n+3)} & \frac{(n+2)(2n+5)(16n^2+81n+90)}{2(n+1)(2n+3)} \\ \frac{96n^4+780n^3+2384n^2+3273n+1723}{(n+1)(n+2)(n+3)(2n+3)(2n+5)} & \frac{68n^3+398n^2+778n+523}{2(n+1)(n+2)(2n+3)} & \frac{96n^4+884n^3+2970n^2+4360n+2403}{4(n+1)(n+2)(2n+3)} \\ \frac{-5(24n^2+117n+143)}{(n+1)(n+2)(n+3)(2n+3)(2n+5)} & \frac{-5(16n+41)}{2(n+1)(n+2)(2n+3)} & \frac{8n^3-44n^2-478n-801}{4(n+1)(n+2)(2n+3)} \end{pmatrix}.$$

Lemma 6 (Exact contiguous trajectory). *For every integer $n \geq 0$,*

$$Y_{n+1} = Y_n T(n).$$

Proof. In the Barnes integrand for F_n , the n -dependent factor is

$$\frac{\Gamma(n+2+s)^2}{\Gamma(n+\frac{5}{2}-s)\Gamma(n+3-s)}.$$

Its quotient after $n \mapsto n+1$ is

$$\frac{(n+2+s)^2}{(n+\frac{5}{2}-s)(n+3-s)}.$$

Apply this identity also after multiplication by s and s^2 . Gamma recurrence and the cubic reduction (6) reduce the three resulting Barnes integrals to the basis $1, s, s^2$; the three coefficient columns are exactly the displayed columns of $T(n)$. All operations are identities in $\mathbb{Q}(n, s)$ after the gamma factors are cancelled. They are first valid on a Barnes contour for $0 < z < 1$; gamma decay permits contour shifting and differentiation. Dominated passage to $z = 1$ gives the assertion. Thus this is an exact contiguous calculation, not a check of finitely many values. $\square$

Put

$$\sigma(n) = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2. \quad (7)$$

Lemma 7 (Inverse-transpose certificate). *For the printed matrix $M(n)$ and the displayed $T(n)$,*

$$T(n)^T M(n) = \sigma(n) I_3. \quad (8)$$

Consequently, if $\Phi_T(N) = T(0) \cdots T(N-1)$ and $H_N = \prod_{m=0}^{N-1} \sigma(m)$, then

$$\mathcal{M}_N = H_N \Phi_T(N)^{-T}, \quad H_N = (-16)^N (2)_N^2 (3)_N^2 (\frac{5}{2})_N (\frac{7}{2})_N^2. \quad (9)$$

Proof. Substitution of the nine displayed rational functions and the nine polynomials defining $M(n)$ gives (8) after clearing the nonzero denominator $(n+1)(n+2)(n+3)(2n+3)(2n+5)$. This is a nine-entry polynomial identity in $\mathbb{Q}[n]$. Multiplying it for $0 \leq n < N$ gives the first identity in (9); factoring each linear term in (7) gives the stated Pochhammer product. $\square$

4 Companion form and exact connection data

Let $e_1 = (1, 0, 0)^T$ and define the Krylov matrix

$$U(n) = [e_1, T(n)e_1, T(n)T(n+1)e_1]. \quad (10)$$

Lemma 8 (Krylov determinant). *Let*

$$D(n) = 3072n^7 + 61824n^6 + 531184n^5 + 2525522n^4 + 7175793n^3 + 12183785n^2 + 11446123n + 4589916.$$

Then

$$\det U(n) = -\frac{15D(n)}{2(n+1)(n+2)^3(n+3)^2(n+4)(2n+3)^2(2n+5)^2(2n+7)}.$$

In particular, $U(n)$ is invertible for every integer $n \geq 0$.

Proof. Expand the determinant using (10) and factor. Every coefficient of D is positive and all denominator factors are positive for $n \geq 0$. $\square$

Lemma 9 (Companion coboundary). *The matrix*

$$C(n) = U(n)^{-1}T(n)U(n+1)$$

is a row companion matrix, $T(n)U(n+1) = U(n)C(n)$, and

$$C(n) = C_\infty + O(n^{-1}), \quad C_\infty = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -35 \\ 0 & 1 & 35 \end{pmatrix}.$$

Moreover

$$Y_n U(n) = (F_n, F_{n+1}, F_{n+2}). \quad (11)$$

Proof. The first two columns of the coboundary identity follow from the definition of $U(n)$; the last column defines the scalar recurrence, so $C(n)$ is companion. Exact rational simplification gives the displayed limit. Equation (11) follows from Lemma 6. $\square$

Lemma 10 (Spectral gap). *The characteristic polynomial of C_∞ is*

$$(\lambda - 1)(\lambda^2 - 34\lambda + 1).$$

Its roots are

$$\mu = 17 - 12\sqrt{2} = (\sqrt{2} - 1)^4, \quad 1, \quad \mu^{-1} = 17 + 12\sqrt{2}.$$

Thus μ is simple and has strictly smallest modulus. A right eigenvector for μ is

$$r_\mu = (17 + 12\sqrt{2}, -18 - 12\sqrt{2}, 1)^T. \quad (12)$$

Proof. This is direct factorization and multiplication by C_∞ . $\square$

Lemma 11 (All three terminal connections are nonzero). *For each $j = 1, 2, 3$, the vector $U(N)^T e_j$ has nonzero connection with the recessive dual mode in Proposition 4.*

Proof. Exact leading terms of the displayed rational matrix $U(n)$ give

$$\lim_{n \rightarrow \infty} \text{diag}(1, n, n^3)U(n) = L := \begin{pmatrix} 1 & 17 & 577 \\ 0 & 24 & 816 \\ 0 & -30 & -1020 \end{pmatrix}. \quad (13)$$

The three row pairings of L with (12) are

$$96(3 - 2\sqrt{2}), \quad 96(4 - 3\sqrt{2}), \quad 120(-4 + 3\sqrt{2}).$$

None is zero: otherwise $\sqrt{2}$ would be rational. Polynomial row scalings do not change whether the corresponding asymptotic coefficient vanishes. Hence all three terminal vectors satisfy the noncancellation hypothesis. $\square$

Let $\mathbf{F} = (F_0, F_1, F_2)^T$.

Lemma 12 (Barnes initial values). *The initial vector is*

$$\mathbf{F} = \sqrt{\pi} \begin{pmatrix} 150G - 128 \log 2 - \frac{146}{3} \\ \frac{24745}{32}G - \frac{14624}{5} \log 2 - \frac{823511}{33600} \\ \frac{7225281}{32}G - \frac{886784}{5} \log 2 - \frac{2818419551}{33600} \end{pmatrix}. \quad (14)$$

Proof. Insert $n = 0, 1, 2$ in the Barnes integrand in (5). Cancelling the two gamma quotients gives

$$\frac{\Gamma(\frac{1}{2} - s)\Gamma(-s)\Gamma(n + 2 + s)^2}{(\frac{3}{2} - s)_{n+1}(1 - s)_{n+2}}.$$

Residue expansion (first at $z = r$ with $0 < r < 1$ and then with $r \uparrow 1$) reduces each of the three cases by partial fractions to

$$\sum_{k \geq 0} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}, \quad \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} = \log 2, \quad \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} = G,$$

plus finite rational tails. Collecting the coefficients gives (14). This is the exact Barnes-moment evaluation the regulator r also justifies the contour endpoint before dominated passage to $r = 1$. $\square$

Lemma 13 (Initial coboundary). *One has*

$$U(0) = \begin{pmatrix} 1 & \frac{152}{5} & \frac{195477}{175} \\ 0 & \frac{1723}{90} & \frac{1963751}{2800} \\ 0 & -\frac{143}{18} & -\frac{165201}{560} \end{pmatrix}$$

and

$$AU(0)^{-T} = \frac{1}{382493} \begin{pmatrix} 30102216645 & -19896711516 & 525137760 \\ 32863650750 & -21712357800 & 573048000 \end{pmatrix}. \quad (15)$$

Proof. Substitute $n = 0$ in (10); exact inversion and multiplication by A give (15). $\square$

Lemma 14 (Exact Catalan cancellation). *The logarithmic and rational terms cancel exactly:*

$$AU(0)^{-T}\mathbf{F} = 450\sqrt{\pi} \begin{pmatrix} G \\ 1 \end{pmatrix}. \quad (16)$$

Proof. Multiply the rational matrix (15) by (14). The two coefficients of $\log 2$ are zero, the two rational constant terms are zero, and the remaining coefficients are respectively $450G$ and 450 . This is an identity in the three-dimensional $\mathbb{Q}$ -space spanned by $1, \log 2, G$; no decimal recognition is involved. $\square$

5 Exact Delannoy transform

This section repairs the finite-prefix claims in the earlier Delannoy argument. It is included because the transform and its error integral are useful exact identities. The main convergence proof below does not require a sign assertion about the transformed coefficient $f(k)$.

Let

$$\mathcal{B}(N, k) = 2^k \binom{2k}{k} \binom{N}{k} \binom{N+k}{k} \quad (0 \leq k \leq N).$$

Lemma 15 (Inverse Delannoy matrix). *The inverse of the lower-triangular matrix $\mathcal{B}$ is*

$$\beta(k, N) = \frac{(-1)^{k+N} (2N+1) \binom{k}{N}}{2^k \binom{2k}{k} (k+N+1) \binom{k+N}{N}} \quad (0 \leq N \leq k). \quad (17)$$

Proof. We prove $\sum_{N=j}^k \beta(k, N) \mathcal{B}(N, j) = \delta_{kj}$. For $k = j$ the sole product is 1. Suppose $m = k - j > 0$ and put $N = j + r$. After removing a nonzero factor independent of r and using

$$\frac{2j+2r+1}{k+j+r+1} = 2 - \frac{2k+1}{k+j+r+1},$$

the sum becomes

$$2 {}_2F_1(-m, 2j+1; k+j+1; 1) - \frac{2k+1}{k+j+1} {}_2F_1(-m, 2j+1; k+j+2; 1).$$

Chu–Vandermonde evaluates the two hypergeometric terms as

$$\frac{(m)_m}{(k+j+1)_m}, \quad \frac{(m+1)_m}{(k+j+2)_m}.$$

Their ratio is $2(k+j+1)/(2k+1)$, so the displayed difference is zero. This proves the inverse formula for every index, rather than checking a finite square. $\square$

Define the scalar normalization

$$H_N = (-16)^N (2)_N^2 (3)_N^2 \left(\frac{5}{2}\right)_N \left(\frac{7}{2}\right)_N^2$$

and, for the first column only in this section,

$$\hat{P}_N = P_{N,1}/H_N, \quad \hat{Q}_N = Q_{N,1}/H_N.$$

There are unique rational sequences f, g such that

$$\hat{Q}_N = \sum_{k=0}^N f(k) \mathcal{B}(N, k), \quad \hat{P}_N = \sum_{k=0}^N g(k) \mathcal{B}(N, k), \quad (18)$$

namely

$$f(k) = \sum_{N=0}^k \beta(k, N) \hat{Q}_N, \quad g(k) = \sum_{N=0}^k \beta(k, N) \hat{P}_N. \quad (19)$$

Theorem 16 (Holonomic k -recurrences). *The sequences $f(k)$ and $g(k)$ in (19) are P -recursive. This conclusion holds for all k as a consequence of holonomic closure applied to the exact finite sums; it is not based on a guessed operator or verification of a finite prefix.*

Proof. By (9), $\hat{P}_N$ and $\hat{Q}_N$ are coordinates of a fixed row multiplied by the three-dimensional rational first-order system $\Phi_T(N)^{-T}$. Four consecutive coordinate functionals are linearly dependent over $\mathbb{Q}(N)$; clearing denominators gives a polynomial-coefficient scalar recurrence. Thus both sequences are holonomic in N .

The kernel $\beta(k, N)$ is proper hypergeometric: both of its shift quotients are rational functions of (k, N) . Zeilberger's creative telescoping theorem [3], equivalently closure of holonomic functions under definite summation, applied to the finite range $0 \leq N \leq k$ therefore produces polynomial-coefficient recurrences for the two sums (19). The argument proves existence and annihilation identically; no order, degree, or minimality inferred by guessing is asserted. $\square$

Theorem 17 (Effective positivity and recessiveness). *For every $n \geq 0$, $F_n > 0$, and*

$$\lim_{n \rightarrow \infty} F_n^{1/n} = 17 - 12\sqrt{2} = \mu. \quad (20)$$

Consequently (F_n) is the unique recessive solution of the companion recurrence in Lemma 9.

Proof. Two beta integrals, two Laplace integrals, and the duplication formula transform the Barnes integral exactly into

$$F_n = \frac{\sqrt{\pi}}{\Gamma(n+1)\Gamma(n+2)} \int_{(0,1)^2 \times (0,\infty)^2} x^{1/2}(1-x)^n(1-y)^{n+1}u^{n+1}v^{n+1} \times \exp\left(-u-v-2\sqrt{\frac{uv}{xy}}\right) du dv dx dy. \quad (21)$$

Every factor is positive, proving $F_n > 0$ directly for every n .

For the exact root rate put $u = na$, $v = nb$ and write

$$F_n = \frac{\sqrt{\pi} n^{2n+4}}{\Gamma(n+1)\Gamma(n+2)} J_n, \quad J_n = \int_D A e^{n\Phi} dx dy da db,$$

where $D = (0,1)^2 \times (0,\infty)^2$,

$$A = x^{1/2}(1-y)ab$$

and

$$\Phi = \log(1-x) + \log(1-y) + \log a + \log b - a - b - 2\sqrt{ab/(xy)}.$$

Let $p = \sqrt{xy}$ and $r = \sqrt{ab}$. Arithmetic-geometric mean gives

$$\Phi \leq 2\log(1-p) + 2\log r - 2r(1+p^{-1}),$$

with equality only when $x = y$ and $a = b$. Maximizing first over r and then over p yields

$$\Phi + 2 \leq 2\log \frac{p(1-p)}{1+p} \leq 4\log(\sqrt{2}-1) =: \Phi_* + 2.$$

The unique maximizer is

$$x = y = \sqrt{2} - 1, \quad a = b = \frac{\sqrt{2} - 1}{\sqrt{2}}.$$

Moreover

$$Ae^\Phi \leq x^{1/2}(1-x)(1-y)^2 a^2 b^2 e^{-a-b},$$

and the right-hand side is integrable on D . Hence $J_n \leq e^{(n-1)\Phi_*} \int_D Ae^\Phi$. Conversely, on a fixed small neighborhood of the unique maximizer, A is bounded below by a positive constant and $\Phi \geq \Phi_* - \varepsilon$. It follows that $J_n^{1/n} \rightarrow e^{\Phi_*}$; this is the elementary Laplace principle with explicit maximizer, not a finite numerical check. Stirling's formula gives

$$\left(\frac{\sqrt{\pi} n^{2n+4}}{\Gamma(n+1)\Gamma(n+2)} \right)^{1/n} \rightarrow e^2.$$

Therefore

$$\lim F_n^{1/n} = e^{\Phi_*+2} = (\sqrt{2}-1)^4 = 17 - 12\sqrt{2}.$$

Lemma 10 and Proposition 4 now identify (F_n) as the unique recessive solution. $\square$

Theorem 18 (Exact integral for the Delannoy coefficient error). *Put $h(k) = g(k) - Gf(k)$ and*

$$\Psi_k(t) = \sum_{N=0}^k \beta(k, N) \frac{R_{N,1}(t^2)}{H_N}. \quad (22)$$

Then, for every $k \geq 0$,

$$h(k) = - \int_0^1 \frac{-\log t}{1+t^2} \Psi_k(t) dt. \quad (23)$$

The sum in (22) is finite and has rational coefficients; in fact

$$\Psi_k(t) = f(k) - (1+t^2)g(k). \quad (24)$$

Proof. Divide (3) by H_N , multiply by $\beta(k, N)$, and sum over $0 \leq N \leq k$. The sum has only $k+1$ terms, so its interchange with the absolutely convergent integral is automatic. Equations (19) give (23). Applying the same inverse sum to $R_{N,1}(t^2)/H_N = \widehat{Q}_N - (1+t^2)\widehat{P}_N$ gives (24). $\square$

Remark 1. Equation (24) shows why the integral identity alone must not be advertised as a decay estimate: without an independent bound it is an exact rewriting of $h(k)$. The earlier inference from finite cancellation to an all- k $O(8^{-k})$ bound is therefore deleted. The positive integral (21) supplies the effective all-index argument actually needed for the proof.

Remark 2 (Other audit corrections). If the unused raw scalar recurrence is indexed as

$$\sum_{j=0}^3 c_j(n) y_{n+j} = 0,$$

the degree pattern is

$$(28, 21, 14, 7), \quad \deg c_j = 7(4-j),$$

not $\deg c_j = 7(3-j)$. The former modular paragraph has also been removed: the eta product $\eta^2(2\tau)\eta^2(4\tau)$ has formal weight 2, and no modular claim is required here. All endpoint passages used above are made by dominated convergence or by the positive Laplace integral.

6 Limit identification and all three columns

Theorem 19 (Hatted sequences and the three-column bridge). *For every $N \geq 0$ and $j = 1, 2, 3$, define*

$$\hat{P}_{N,j} = \frac{P_{N,j}}{H_N}, \quad \hat{Q}_{N,j} = \frac{Q_{N,j}}{H_N}, \quad H_N = (-16)^N (2)_N^2 (3)_N^2 \left(\frac{5}{2}\right)_N \left(\frac{7}{2}\right)_N^2. \quad (25)$$

Then

$$\frac{P_{N,j}}{Q_{N,j}} = \frac{\hat{P}_{N,j}}{\hat{Q}_{N,j}} \longrightarrow G \quad (j = 1, 2, 3).$$

Proof. By (9), for each standard column e_j ,

$$\begin{pmatrix} \hat{P}_{N,j} \\ \hat{Q}_{N,j} \end{pmatrix} = A\Phi_T(N)^{-T} e_j. \quad (26)$$

Theorem 17 identifies $\mathbf{F}$ as the recessive initial vector. Lemma 11 verifies the nonzero terminal coefficient separately for each of $j = 1, 2, 3$. Proposition 4 and Lemma 14 therefore give

$$[A\Phi_T(N)^{-T} e_j] \longrightarrow [AU(0)^{-T} \mathbf{F}] = \left[450\sqrt{\pi} \begin{pmatrix} G \\ 1 \end{pmatrix} \right] = \left[\begin{pmatrix} G \\ 1 \end{pmatrix} \right].$$

The second coordinate of a normalized representative tends to the nonzero number 1, so $\hat{Q}_{N,j}$ (and hence $Q_{N,j}$) is nonzero for all sufficiently large N , and $\hat{P}_{N,j}/\hat{Q}_{N,j} \rightarrow G$. Finally $H_N \neq 0$, so the equality of the hatted and original ratios in (25) proves all three assertions of Theorem 1. $\square$

References

- [1] S. Weinbaum et al., *On Conservative Matrix Fields: Continuous Asymptotics and Arithmetic*, arXiv:2507.08138, 2025.
- [2] NIST Digital Library of Mathematical Functions, *Meijer G-function*, Section 16.17, <https://dlmf.nist.gov/16.17>.
- [3] D. Zeilberger, *The method of creative telescoping*, J. Symbolic Comput. **11** (1991), 195–204.